



Lucia Piro

PhD student in Plant Physiology

Education

- Sep'18 – Aug'20 **MSc Plant Biotechnology** Wageningen University and Research (NL)
sp. Molecular Plant Breeding and Pathology
Final mark: "cum laude"
Thesis title: Genetic analysis of accumulation of carotenoid-derived volatile compounds in tomato fruit
Skills: GC-MS, HPLC-MS, qPCR, QTL mapping, data visualization
Academic Consultancy training: I worked as group manager with a multicultural team of 6 students to research the use of orchids' molecules in the cosmetics and food industry.
Thesis grade: 9/10
- Oct'15 – Jul'18 **Bachelor's degree in Agri-food Biotechnology** Università degli Studi di Milano (IT)
Final mark: 110 cum laude/110
Thesis title: Identification of stomata-specific genes in *Arabidopsis thaliana*
skills: genotyping, qPCR, GUS assays
- Sep'10 – Jul'15 **Scientific Highschool** Liceo Scientifico Marie Curie Tradate (IT)

Personal information

Day of birth: 26-12-96

Driving licence: B

Contact

Address:

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8050, Zurich
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Languages

Italian (mother tongue)

English (C1)

Programs

Ms. Office package, R,
Affinity Designer, LaTeX

Interests

Drawing, art, cooking and
nutrition

Work experience

- Oct'20 – Jun'25 **PhD Candidate** ETH Zürich, Department of Environmental System Science
Thesis topic: Unravelling the role of carbohydrates in stomatal immunity
Skills: Western blots, qPCR, PCR, plant physiology and biochemistry, confocal microscopy, starch staining, protoplasts isolation, starch and sugars quantification via enzymatic assays, data visualization with R
Courses: Project management for research, Plant disease diagnostics, Statistics in functional genomics, Design course for scientists, Film making for scientists
- Feb – Aug'20 **Global Agriculture Intern** McCain Food Limited, Canada
Project title: Survey of Potato Mop Top Virus (PMTV) in Manitoba (Canada).
I was admitted to the McCain internship program where I experienced six months understanding and solving PMTV spread in the Manitoba praries

Relevant activities

- Aug '23 **Conference talk** ICAR 2023 - Japan
Title: Investigating the role of Carbohydrates in Bacterial-triggered stomatal movements
Highlight: Short-talk award
- Dec'22 **Plant Science Centre Symposium organisation** ETH Zürich
Title: From place to space - Tracing the spatial dimension of plant sciences
Tasks: helping with the catering and label tags, website design, speaker choice and hospitality, presenting the speakers

Oct'22 – Mar'23 **Bachelor student supervision**

ETH Zürich

Report title: Development of a silencing strategy in Arabidopsis thaliana's guard cells

Oct'21 – Mar'25 **VMUSYS, president and head of communication**

ETH Zürich

Tasks: VMUSYS is the scientific staff association at D-USYS (ETH). I am here compiling and sending the monthly newsletter to department, organising meetings with the team and brainstorming further steps for improving the association network.

Publications

Feb'23

Plant Signalling and behaviour

Title: Arabidopsis Sucrose Synthase 3 (SUS3) regulates starch accumulation in guard cells at the end of day

Piro L., Flütsch S., Santelia D.

Mar'24

Plant Physiology, in revision

Title: Guard cell starch metabolism at the intersection of environmental stimuli and stomatal movements

Dang T., **Piro L.**, Pasini C., Santelia D.

Highlight: Figures design via Affinity 2

2024

Aim: Plant Physiology, writing

Title: Malate and starch metabolism participate to low CO₂ triggered-stomatal opening

Piro L., Silva-Alvim F., Santelia D., Blatt M.

2024

Writing, finalising experiments

Topic: The role of carbohydrate metabolism in stomata under pathogen attack

Piro L., Santelia D.